

THE UNIVERSITY OF
SYDNEY

Tutorial Class	
Student Name	
SID	

CONFIDENTIAL QUIZ PAPER

This paper should not be removed from the quiz venue.

COMP5339 – Data Engineering

Tutorial Quiz – 2
Individual Quiz (5%)

Quiz Writing Time: **30 minutes**
Thursday, 21 May

Instructions to Students

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered.**
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor before leaving the room.

Part I – Multiple Choice Questions

Question 1: This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

A. From a data engineering perspective, what is the main extra requirement for a supervised learning pipeline compared with an unsupervised learning pipeline?

- ☐ It must always use image data
- ☐ It does not require preprocessing
- ☐ It only works on structured data
- ☐ **It must include ground-truth labels linked correctly to features**

B. Why is metadata analysis attractive in image pipelines?

- ☐ A. It is always more accurate than image content
- ☐ **It can provide useful structured information without full image-content analysis**
- ☐ It avoids all storage requirements
- ☐ It replaces feature extraction completely in every case

C. Which of the following is an example of a transactional data stream?

- ☐ **Credit card purchases by customers**
- ☐ Temperature readings from sensors
- ☐ Road traffic speed measurements
- ☐ Weather history reports

Part II – Short Answer Questions

Provide answers in the provided boxes.

Question 1: What is the main purpose of tokenisation in text preprocessing? What are the storage options for storing the output of tokenisation? (2 Marks)

Tokenisation mainly splits raw text into smaller units such as words, subwords, or terms so the text can be processed and converted into features for analysis or machine learning.

Storage options include:

- As text/JSON documents for flexible storage of token lists
- In relational tables with one row per document or one row per token
- As sparse matrices / document-term matrices when preparing features for ML

Question 2: Why is a traditional DBMS not sufficient for many stream processing applications? (2 Marks)

A traditional DBMS is mainly designed for data at rest, where queries are executed over stored data. Stream applications require continuous processing of live data with low delay, which a DBMS does not handle as naturally as a stream processing system.

Scenario 2: Explain the difference between scale-up and scale-out in scalable data engineering. Why is scale-out generally preferred for big data systems? **(3 Marks)**

Scale-up means increasing the power of a single machine by adding more CPU, memory, or storage. Scale-out means adding more machines or nodes to a cluster. Scale-out is generally preferred for big data systems because a single server has physical and cost limits, while clusters can grow more easily and support shared-nothing distributed processing.

EXTRA SHEET